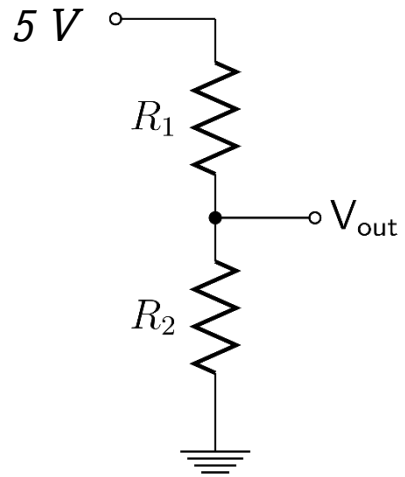
Embedded projects 1
Weekly exercises 1

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Exercise 1

The tasks are submitted by showing the final result to the teacher.
Be prepared to explain how your code works!



Task 1:

- Find a partner and install Arduino IDE on both computers if it's not already installed.
- Gather the following components: Arduino Mega, programming cable, breadboard, light sensor (LDR), and a ~1.8 kΩ resistor.

Task 2:

- Implement the circuit and Arduino code to measure the strength of the *analog* signal from the sensor (V_{OUT}).
- LDR replaces the R1 resistor in the circuit.



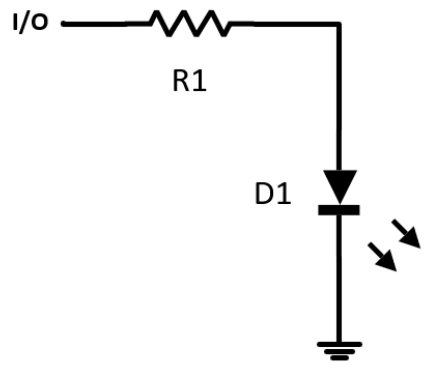
Task 3:

- Convert the measured ADC result into voltage.
- Print the result to the serial monitor with the unit (V).
- The measurement and printing frequency should not exceed 1 Hz.

Exercise 2

Task 1: Add an LED (red or green) and a ~270Ω resistor into arduino digital output

Task 2: Modify your code such that if you cover the LDR, the LED lights up and turns off after uncovering the LDR





Exercise 3 – Motor encoder operation

Tampere University of Applied Sciences

By using an oscilloscope, test that your lego motor encA & encB pins outputs pulses

Step 1: Connect ethernet PCB into the lego motor
Step 2: Connect power supply into the ethernet board
Step 3: Connect oscilloscope into the PCB, hook into the GND and head first into the ENCA
Step 4: Repeat for ENCB
Step 5: Repeat for other ethernet connector, cable, & motor

!!! Do not remove the ethernet cables from the PCB!!!

When connected to a LEGO motor, these cables have the following pinout:



Pin	Description
1	motor –
2	motor +
3	GND
4	5V
5	quadrature encoder A
6	quadrature encoder B



